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ORIGINAL RESEARCH

An improved air quality and climate control monitoring system using fuzzy logic for enclosed areas

Anas Bushnag[1]

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Abstract
Indoor pollution is significant for many people since it has greater effects than outdoor pollution. Maintaining a healthy indoor environment is critical. Therefore, indoor environments are ventilated to reduce health risks and improve air quality. In these kinds of situations, automation to refresh surrounding environments is important because of the current fast-paced lifestyles of most people. The technique proposed in this paper automates and manages indoor temperature, humidity, and air quality. This is achieved by using an Arduino microcontroller with fuzzy logic integration. The air quality is measured by using an MQ-135 sensor, whereas the temperature and humidity of the indoor environment are tracked by using a DHT-11 sensor. These sensors transmit their readings to a microcontroller to set the ventilation speed for the indoor environment, which removes harmful gases and maintains acceptable temperature and humidity levels. The performance of the proposed system shows excellent results in controlling and monitoring air quality, temperature, and humidity compared with a previous version due to an increased range between the minimum and maximum values of the ventilation speed. Moreover, the ventilation DC motor of the fan in the proposed system runs at low speeds when necessary, which leads to lower power consumption compared to the previous version. This reduction is due to the membership function added to the ventilation speed of the fuzzy controller, which adapts the working time interval and room ventilation speed. A comparison of the new proposed system and previous version is conducted to identify the differences between the two and to highlight the effectiveness of the new model.

Keywords AQI · Air quality · Fuzzy logic · Ventilation

1 Introduction

Many people spend most of their time in enclosed environments (Kankaria et al. 2014; Son and Son 2019; Ho et al. 2020). This increases the importance of indoor air quality, which usually has lower air quality index (AQI) values than open environments (Pradityo and Surantha 2019). The probability of people acquiring and transmitting diseases is much higher indoors than outdoors (Gupta et al. 2015). People who suffer from breathing difficulties and asthma are the most exposed to air pollution (Saini et al. 2020; Preethi and Tamilarasan 2021). Hence, air quality management and control are required (Said et al. 2019; Sun et al. 2017; Yi et al. 2015).

In this paper, the AQI will be used as a metric for measuring how much pollution is in the air (Chakma et al. 2017; Grace and Manju 2019). The AQI system works by measuring the quality of the surrounding environment in the range of 0 to 500 degrees by using a thermometer (Zhou et al. 2020). This system is used to show the volume of pollution in the air (Aggarwal et al. 2017). Atmospheric air contains oxygen, nitrogen, and smaller amounts of several other gases. The AQI measures the five main components of air pollution: ground-level ozone, carbon monoxide, sulfur dioxide, nitrogen dioxide, and airborne particles (Kang and Qu 2017). Of these five factors, airborne particles and ground-level ozone are the most important since they can harm human health. Chemical emissions such as nitrogen dioxide, carbon monoxide, and methane, which are produced by industrial facilities, car exhaust, gasoline vapors, and many other sources, interact with sunlight, resulting in the production of ground-level ozone. This has the opposite effect of high-level ozone that protects people

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from the sun's harmful radiation. There are two types of airborne particles in air: solid and liquid droplets. Usually, car exhaust systems, volcanoes, smokestacks, wildfires, construction sites, and many other sources create particles. Air is not considered to be polluted if the AQI value is under 50 degrees. The air quality is considered acceptable for an AQI between 51 and 100 degrees. The air quality is considered polluted and unhealthy if the AQI exceeds 100 degrees.

Fuzzy logic systems work to convert words into numbers in an attempt to emulate human thinking (Ganesh et al. 2017). A normal variable consists of only two parameters: name and value. In fuzzy logic, a linguistic variable is used. A linguistic variable is more complicated and has five parameters: variable name, a set of terms, universe of discourse, syntax rules, and semantic rules. It can be defined as follows:

$$LinguisticVariable = <X, T(X), U, G, M>$$

Fuzzy logic consists of three phases: fuzzification, inference, and defuzzification. In the fuzzification phase, the system processes the inputs by performing coding on these inputs. The inference phase defines a set of rules by utilizing a base rule to control the system's behavior. Eventually, the defuzzification phase decodes the processed data to provide the output area. The output area is converted into a numeric scaler by using one of the defuzzification methods. The output scaler will be the actual output of the fuzzy logic controller.

The proposed technique is used to design a system whose behavior is managed by an Arduino microcontroller and fuzzy logic. Initially, the attached sensors measure the temperature, humidity, and air quality. The system converts the analog signals from these sensors into digital measurements. The digital measurements are passed to the fuzzy logic controller. These measurements are processed by the fuzzy logic controller in three phases: fuzzification, inference, and defuzzification. The defuzzification scalar output is sent to the Arduino system to adapt the DC motor fan in terms of ventilation speed. In this system, a DC motor is used for demonstration purposes. Nevertheless, the ventilation mechanics can involve an air conditioner, a fan, or any other ventilation mechanism system. A set of rules is designed in the inference stage to control the ventilation system speed. These rules determine the system behavior output based on the measurements obtained from the sensors. This technique is an improved version of the technique proposed in a previous paper (Bushnag 2020). Therefore, a new membership function is added, and some of the rules are changed accordingly. A MATLAB R2015b fuzzy logic toolbox is used to develop the new technique for the fuzzy logic controller. Moreover, a comparison of the new technique and previous technique (Bushnag 2020) is conducted to highlight the differences between the two.

This paper is organized as follows: In Sect. 2, the related work is presented. In Sect. 3, the proposed system model is discussed, which contains the circuit design, fuzzy logic controller, and defuzzification method. In Sect. 4, the results and analysis are shown. Finally, the conclusions are presented in Sect. 5.

2 Related work

There are many indoor air quality control and monitoring scheme research papers in the literature. Some of these research papers are discussed below to present their motivations and objectives.

In Pradityo's system (Pradityo and Surantha 2019), an IoT indoor air quality control and monitoring technique is developed and connected to an IoT server. The proposal consists of four stages: the enclosed area, fuzzy logic stage, mechanical system, and IoT server. The environment of the room contains two sensors: CO_2 and PM10. Moreover, the room contains an exhaust fan. In the fuzzy logic process, the settings for the exhaust fan operation intervals are automated. This is achieved by adding fuzzy rules to control the gas intensities. The final defuzzification output is a value that decides how much time the exhaust fan will operate for to improve air quality in the surrounding environment. On the cloud server, information is obtained from the PM10 and CO_2 sensors and then sent to the AWS, which has a dashboard for system monitoring. The dashboard presents real-time graphs for PM10 and CO_2 values. System users have the ability to access all the data on the cloud server using the internet wherever they are. However, the air quality measurements for this proposed method are based on only two factors that are determined by PM10 and CO_2 sensors. This might affect the system's accuracy since more sensors should be added to improve the overall accuracy of the system.

In Kumber's paper (Kumar and Jasuja 2017), indoor air quality was also investigated. Their proposed system is connected to a cloud server. The system contains many sensors that are used to observe different factors of air quality: examples include how much carbon monoxide, how much carbon dioxide, PM2.5 level, air pressure, and humidity. The data collected from these sensors are transmitted to the cloud server. On the cloud server, all uploaded data are graphically represented for a better user experience. The system is built using a Raspberry Pi that is connected to the cloud server throughout the MQTT protocol. This is necessary to ensure a stable connection between the Raspberry Pi and cloud server. An access point is used to connect the proposed indoor monitoring

technique to the internet via Wi-Fi. This system contains more measuring sensors than Pradityo's system, which is considered better in terms of the air quality accuracy level. However, the proposed system has no technique to overcome the low air quality of indoor environments.

Sharma's proposed system (Kumari et al. 2013) uses two-way ducting integrated with a fuzzy logic controller. The main goal of this system is to distribute the air equally in indoor environments. The fuzzy logic controller inputs of the system are humidity and temperature. The fuzzy logic controller outputs of the system are the pump, fan speed, and fan duct. The proposed system implements the Mamdani fuzzy model. The model consists of 30 rules to manipulate the system output. Moreover, the fuzzy logic controller manages the room's humidity and temperature. In contrast, the system has no air quality sensor. This will not allow users to monitor the air quality of the enclosed environment.

In Mukesh's system (Mukesh and Akula 2017), two sensors are used: CO_2 and CO. The proposed system uses a Raspberry Pi 3 to monitor indoor air quality. The ultimate system decision is to turn the fan DC motor on/off. This is controlled via manually pressing the on/off button on a smartphone application. The system defines a threshold for the air quality level. If the air quality level exceeds the threshold, the system recommends that the user turn on the ventilation fan via the smartphone application and vice versa. This paper suggests some improvements for the system in the future. One is to add more sensors to improve the accuracy and reliability of the system. The main issue with this technique is automation, since users must activate and deactivate the ventilation fan themselves. This is the main disadvantage of the technique, which can be improved upon in the future.

In Leadekar's system, (Ladekar and Daruwala 2019), the indoor air quality is estimated using different sensors. The collected data are uploaded and stored in the cloud. The real-time data are visualized using graphs for a better experience. The system user is notified when the air quality of the surrounding environment is poor. The proposed system uses a universal asynchronous receiver/transmitter (UART) to control the serial device interfacing. This is achieved by converting the parallel data into a serial bitstream. In this system, the MQTT protocol is used to publish and subscribe to data on the cloud server. The system utilizes Amazon Web Services (AWS) because of the following: scalability, flexible capacity, speed, agility, global reachability, security, lack of upfront investment, and project focus. The system design is based on a Raspberry Pi that is connected to several sensors. These sensors measure CO_2, O_2, and dust. Moreover, the system displays the air quality information on a LCD screen for greater usability. However, the system measures neither temperature nor humidity for the indoor environment. Additionally, it does not provide any solution for ventilation apart from a warning message.

The system developed by Nasution in (Nasution et al. 2020) uses an ESP32 as a controller that is connected with various sensors for air quality measurement. The system can be used to monitor temperature, humidity, dust particles, and gases. ESP32 was chosen because it is a low-cost and low-power device that has built-in WiFi and Bluetooth. BME280 measures the surrounding temperature and humidity. MQ-136 senses many types of gases such as NH3, CO, and NO2. A Winsen ZE03-SO2 sensor was used to estimate SO2 levels in the indoor environment. All of the information provided by the previous sensors is presented on a LCD display to show the current conditions of an indoor environment in terms of air quality. However, the system does not provide any solution for mitigating poor air quality.

3 Proposed system model

The proposed system architecture includes five steps: sensor measurements, fuzzification, inference, defuzzification, and mechanical control. Figure 1 shows the proposed system diagram. In the first step, sensors sense the environment and convert the analog signals into reading values. These values are the fuzzification step inputs, which are processed by applying coding to them. In the inference step, a set of rules is applied to shape the proposed system behavior. In the defuzzification step, the values are decoded depending on the selected rules created in the previous step. The output of the fuzzy logic controller is a surface area. This surface is then converted into a scalar that will be the actual output of the system. In the last step, mechanical output, such as a ventilation fan or air conditioner, is fed by the scaler from the defuzzification step to control the speed. The mechanical output has two features: The first feature is to turn on/off the mechanical system, and the second feature is to set the velocity of the mechanical system. In this study, a simple hardware system design is developed, and

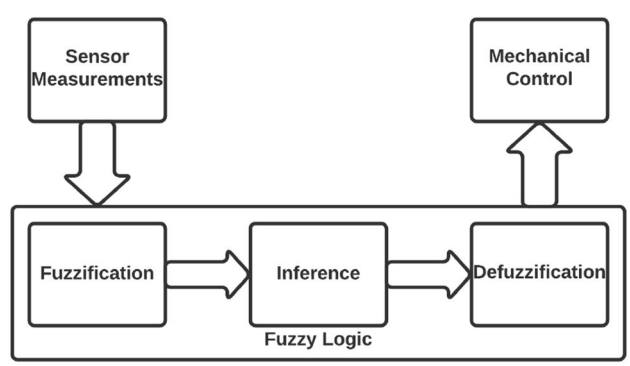

Fig. 1 The proposed system diagram

a more complicated controller system design is acquired. This improves the system reliability and decreases the system power consumption. In the first subsection, an Arduino system design is proposed. In the second subsection, a fuzzy logic controller is developed. In the third subsection, the implemented defuzzification method is presented.

3.1 Circuit design

The design of the system hardware includes three sensors, an Arduino microcontroller, and a DC motor. DHT-11 sensors are used in the system for measuring the temperature and humidity and a MQ-135 sensor is used to monitor the air quality. The MQ-135 senses smoke, alcohol, CO_2, benzene, sulfide, ammonia, and several other pollutants. The MQ-135 sensor is used to estimate the air quality of the surrounding area. On the other hand, DHT-11 works for measuring the temperature and humidity. The sensor data are fed to the system after processing by the fuzzy logic controller. Finally, the system supplies the correct amount of current to the DC motor of the ventilation fan based on the conditions of the surrounding environment. The design of the system is shown in Fig. 2.

3.2 Fuzzy logic controller

Using fuzzy logic in the proposed system provides extreme flexibility and adaptability for the ventilation motor speed. Instead of having a switch with only 0 and 1 values, a great degree of freedom is afforded by the selection of the exact speed of the ventilation motor based on the current situation in the environment. This will lead to a more efficient system with less power consumption.

There is no exact method for the selection of membership functions. However, the distribution of the data can help to select the most appropriate membership function. Moreover, using a trial and error method can also be beneficial. Applying different shapes of membership functions might not affect the system's accuracy but will determine the computational time. In the proposed system, the triangular membership function is utilized for the data with obvious minimum and maximum values in real life, such as humidity. On the other hand, parameters such as temperature have no exact minimum or maximum values. Therefore, a trapezoidal membership function was selected.

The fuzzy logic controller proposal includes three inputs: temperature, humidity, and air quality. There is only one output: ventilation speed. The temperature range is set to be within the range of – 10 and 40 degrees. The range of humidity is set to be within the range of 0% and 100%. The range of air quality is set to be within the range of 0 and 500. The range of the output, which is the speed of the DC motor, is set to be between 0 and 255. The output range is selected according to the pulse width modulation (PWM) of the microcontroller. This is used to manipulate the ventilation speed. The fuzzification step represents the temperature by using five-member trapezoidal functions: very cold, cold, normal, warm, and hot. Figure 3 shows the temperature representation. The fuzzification step represents the humidity using five-member triangular functions: very low, low, medium, high, and very high. Figure 4 shows the humidity representation. The fuzzification step represents the air quality using five-member trapezoidal functions: good, moderate, polluted, very polluted, and very unhealthy. Figure 5 shows the air quality representation. In the defuzzification step, the ventilation speed is represented by two-member trapezoidal functions and three-member triangular functions: very slow, slow, normal, high, and very high. Figure 6

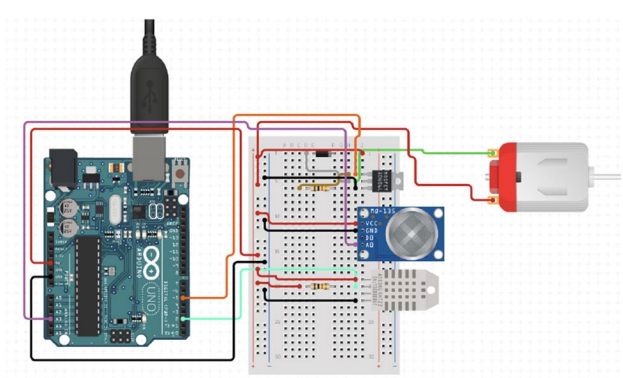

Fig. 2 Arduino circuit design

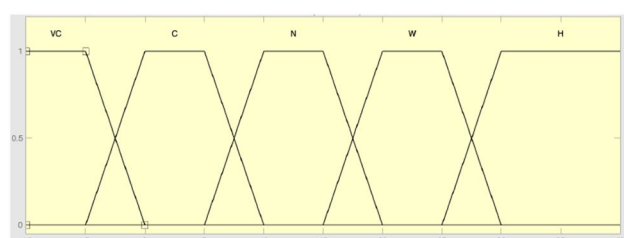

Fig. 3 Member functions of temperature

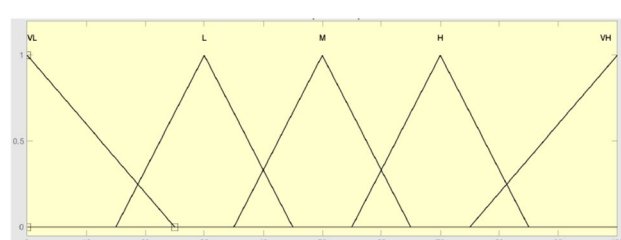

Fig. 4 Member functions of humidity

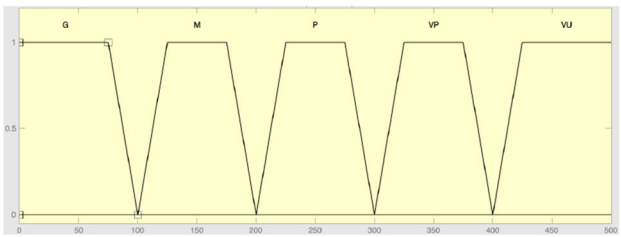

Fig. 5 Member functions of air quality

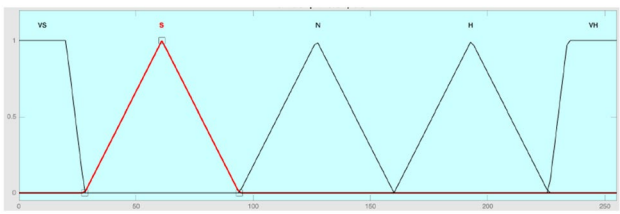

Fig. 6 Member functions of ventilation speed

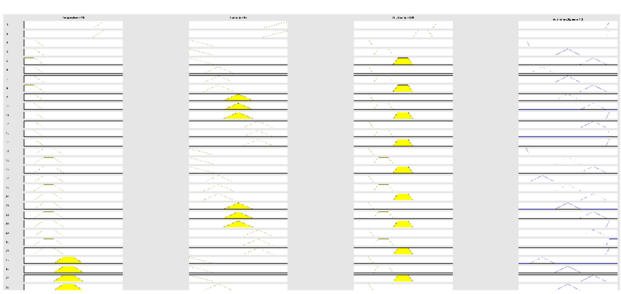

Fig. 7 The set of rules used by the proposed system fuzzy logic

shows the ventilation speed representation. A one member function is added to make the total number of membership functions five instead of four as in the previous version (Bushnag 2020). This improves the accuracy of the proposed system because the surface area of these memberships is smaller than before. The center of gravity calculation will be affected in a more precise way because there is less area to cover. Moreover, the range between the minimum and maximum values is increased to produce a much more accurate output for the ventilation speed. The two edge membership functions are trapezoidal, and the three middle functions are triangular. These triangular membership functions provide a much-distinguished output based on the input values.

The fuzzy logic controller consists of 50 rules to manage the behavior of the system. Figure 7 presents some of these rules. In the new proposed system, two of the rules are changed to accommodate the new ventilation speed membership functions. These rules in the previous version were as follows (Bushnag 2020):

- If the temperature is *very cold*, humidity is *very low*, and air quality is *good*, the ventilation speed will be *slow*.
- If the temperature is *cold*, humidity is *very low*, and air quality is *good*, the ventilation speed will be *slow*.

They now become as follows:

- If the temperature is *very cold*, humidity is *very low*, and air quality is *good*, the ventilation speed will be *very slow*.
- If the temperature is *cold*, humidity is *very low*, and air quality is *good*, the ventilation speed will be *very slow*.

This will allow the system to operate the ventilation motor at very low speeds when it is necessary, unlike the previous version (Bushnag 2020), which reduces the power consumption.

3.3 Defuzzification method

In the proposed model, the center of gravity (COG), also known as the centroid of area (COA), is used. This method provides a numeric scale based on the fuzzy set center of gravity (Allahviranloo and Saneifard 2012). It divides the total area of membership action that represents the member function into subareas. In each subarea, the area and the center of gravity are calculated. The defuzzification value is determined by summing these subareas. The following equation is used to calculate the defuzzification value:

$$X = \frac{\sum_{i=1}^{n} A_i x_i}{\sum_{i=1}^{n} A_i} \tag{1}$$

where X is the defuzzification value, n is the number of subareas, A_i is the area of the current subarea, and x_i is the center of gravity of the current subarea.

4 Results and analysis

Two separate scenarios are demonstrated to highlight the differences in the response of the system for each situation. Each scenario will have different outputs. The system simulation will be tested under different environmental circumstances. The results and analysis for each scenario are presented to analyze the response of the system.

4.1 Scenario 1: when sensors read high values

In the proposed fuzzy logic controller, the two first rules function for handling high-value readings from sensors. Consequently, when one of the three sensors for temperature, humidity, or air quality reads a high value, the ventilation

speed will be at its maximum to overcome the pollution in the enclosed area.

The relationship between the air quality and ventilation speed is shown in Fig. 8. In the case that the range of the air quality is between 0 and 175, the ventilation speed should be approximately 193 instead of 144 as in the previous version (Bushnag 2020) because the air quality is somewhere between good and unhealthy for asthmatic people. Nevertheless, when the AQI surpasses value of 175, the ventilation speed ranges between 240 and 243, which is considered a dramatic increase. These values replace those of 220 and 224 in the previous version. The main reason for this rapid increase is to reduce air pollution in the surrounding environment. The relationship between air quality and ventilation speed is not always linear because other factors are involved, such as the current temperature and humidity. The newly proposed version provides more ventilation than the previous version to create a healthier environment.

The relationship between the temperature and ventilation speed is shown in Fig. 9. The ventilation speed ranges between 242 and 243 instead of 220 and 224, as in the previous version (Bushnag 2020). The main reason for this is that in some cases, even when the temperature sensor reading is low, the ventilation speed output has a high value. This is because of the other two elements: humidity and air quality. Note that the range of this relationship is increased due to the added ventilation speed member function.

The relationship between the humidity and ventilation speed is demonstrated in Fig. 10. When the humidity is less than 35%, the ventilation speed will vary between 193 and 195 instead of 145 and 150, as in the previous version (Bushnag 2020). If the humidity exceeds 35%, the range of the ventilation speed will rapidly increase to a value between 240 and 243 instead of 200 and 224, as in the previous version. Note that the ventilation speed does not fall below 193 because of the other two elements. Additionally, the ventilation speed varies between 240 and 243 for the same reason.

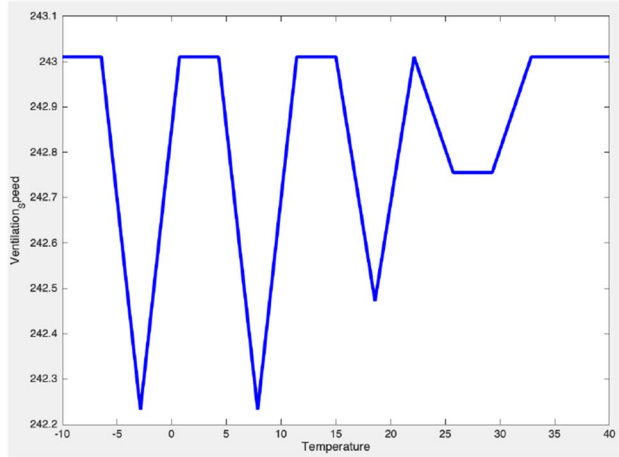

Fig. 9 Temperature vs. ventilation speed

4.2 Scenario 2: when the sensors have different values

There are many cases in which the sensors will have different readings. For instance, a sensor can read a medium-high value, whereas the second sensor can read a low value and the third sensor can read a medium-low value. In this situation, the system should provide a suitable output based on the current sensor values. This is the main responsibility of the other 48 rules of the fuzzy logic controller system. In this subsection, some of these situations are selected to demonstrate the system response in various conditions. In the first test, the temperature, humidity, and air quality are very cold, very low, and good, respectively. The system rule

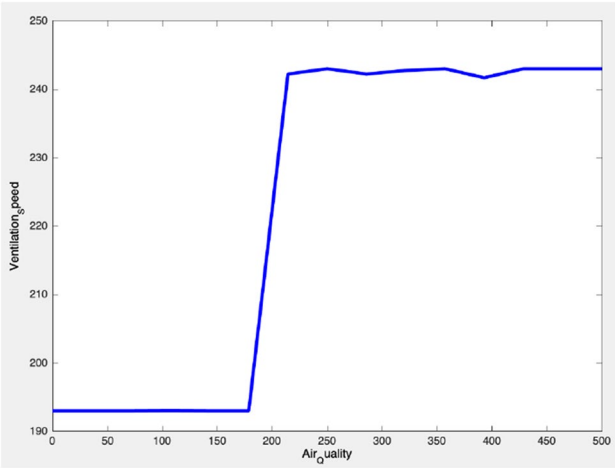

Fig. 8 Air quality vs. ventilation speed

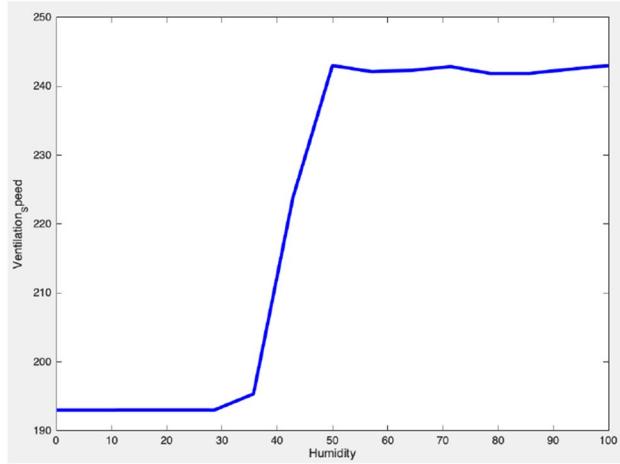

Fig. 10 Humidity vs. ventilation speed

for this behavior suggests that the ventilation speed should be very slow. In another test, the temperature was cold, the humidity was medium, and the air pollution was good. The system rule for this behavior suggests that the ventilation speed needs to be normal. Hence, to understand the system behavior, the relationship between all inputs must be disclosed. To disclose the triple relationship, three double relationships are obtained: temperature and humidity, temperature and air quality, and humidity and air quality. These different relationships can define exactly how the system behavior works.

The relationship between temperature and humidity on one side and the ventilation speed as the output on the other side is demonstrated in Fig. 11. When the temperature falls below 20 degrees and the humidity decreases below 35%, the ventilation speed will be approximately 100 because of the normal surrounding environmental conditions. On the other hand, when the temperature exceeds 25 degrees or the humidity exceeds 35%, the ventilation speed range will increase dramatically to between 240 and 243 instead of 215 and 224, as in the previous version (Bushnag 2020). The yellow surface looks smoother compared to the previous version because the range difference between the minimum value and maximum value of the ventilation at high speeds is lower in the new proposed system.

The relationship between temperature and air quality on one side and the ventilation speed as the output on the other side is demonstrated in Fig. 12. If the temperature is below four and the air quality is below 70, the ventilation speed is 120 instead of 80, as in the previous version (Bushnag 2020). This is because good air quality and low temperature reduce the need for high-speed ventilation. On the other hand, when the temperature range is between 10 and 30 and the air quality ranges between 100 and 200, the ventilation speed can range between 180 and 230 instead of 120 and 190, as in the previous version. The main reason for this increase is the high values of temperature and air pollution. Finally, if the temperature exceeds 35 degrees or the air quality exceeds 250 degrees, the ventilation speed operates between 240 and 243 degrees, which is considered the maximum range. This is an attempt by the system to reduce air pollution or temperature. As noted, the newly proposed method has a higher ventilation speed than the previous method due to the added ventilation speed membership function.

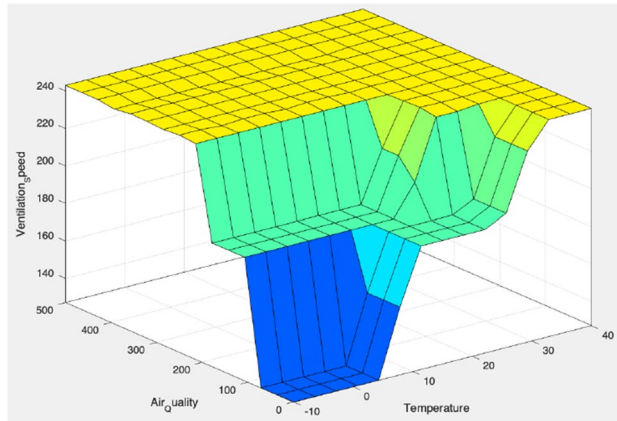

Fig. 12 Temperature vs. air quality vs. ventilation speed

Figure 13 demonstrates the relationship between air quality and humidity on one side and the ventilation speed as the output on the other side. This relationship has four stages, which are represented by different colors. When the AQI is below 80 and the humidity is below 17%, the ventilation speed will be close to zero because of the normal condition of the surrounding environment. If the air quality value ranges between 100 and 200 and if the humidity value range is between 20% and 40%, the ventilation speed will range between 100 and 160 instead of 50 and 120, as in the previous version (Bushnag 2020). When the AQI is between

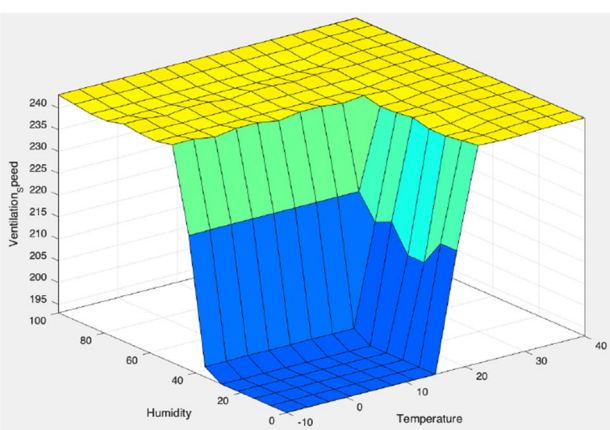

Fig. 11 Temperature vs. humidity vs. ventilation speed

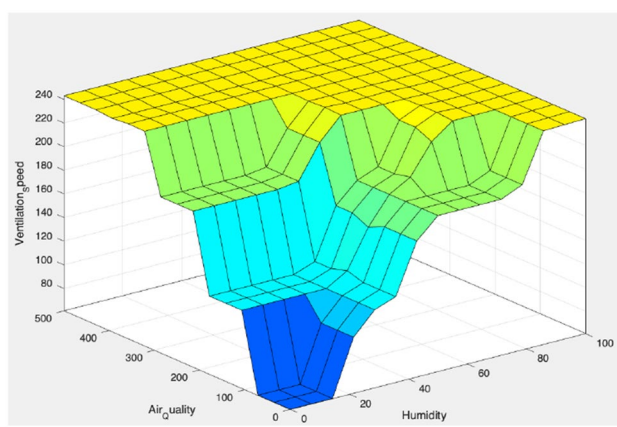

Fig. 13 Humidity vs. air quality vs. ventilation speed

Table 1 Performance Comparison of the ventilation speed

Technique	Max	Min	Range
Previous version (Bushnag 2020)	224	20.5	203.5
The proposed system	243	11.4	231.6

Table 2 Three examples to show the different responses between the proposed system and the previous version

Parameter	Ex.1	Ex.2	Ex.3
Temperature	0	10	20
Humidity	0	30	50
Air quality	0	150	250
Previous version (Bushnag 2020)	20.5	80	224
The proposed system	11.4	127	243

250 and 350 and the humidity is between 50 and 85%, the ventilation speed will be between 180 and 230 instead of 120 and 200, as in the previous version. This is because of the poor surrounding environmental conditions. Finally, when the AQI is higher than 350 or the humidity is higher than 85%, the ventilation speed will be approximately 243 instead of 224, as in the previous version, which is considered the maximum rate. The proposed system always yields a higher ventilation speed compared to the previous version in these types of situations.

4.3 The proposed system comparisons

4.3.1 Proposed system vs. previous version

In this subsection, a comparison between the proposed system and previous version (Bushnag 2020) is conducted to show the differences and how the new system provides more accurate results compared to the previous version.

After adding the new membership function to the ventilation speed output, the surface area that needs to be covered by the output membership functions is reduced. This is because of the use of five membership functions instead of four. As shown in Table 1, the maximum value of the system is changed from 224 to 243, and the minimum value of the system is decreased from 20.5 to 11.4. Because of the new membership function added to the defuzzification stage, the range is increased from 203.5 to 231.6, which improves the distribution of the ventilation speed values. This allows the system to operate at either low or high speeds when necessary. Table 2 shows the different responses of the proposed system compared to the previous version (Bushnag 2020) based on the parameter values.

4.3.2 Proposed system vs. (Preethi and Tamilarasan 2021)

A comparison between the proposed system and technique introduced in (Preethi and Tamilarasan 2021) in terms of hardware, measurements, analysis method, environment, and system output is conducted. The proposed system uses a DHT-11 to measure the temperature and humidity, whereas (Preethi and Tamilarasan 2021) uses the DHT-22 to measure only temperature. The proposed system uses an MQ-135 sensor to monitor the air quality. On the other hand, the technique in (Preethi and Tamilarasan 2021) uses an MQ-7 for monitoring carbon monoxide. However, both systems use an Arduino Uno as a microcontroller for data processing. The proposed system uses fuzzy logic, and it is suitable for indoor environments, unlike (Preethi and Tamilarasan 2021), which uses an improved-adaptive neuro-fuzzy inference system (I-ANFIS) that is suitable for outdoor environments. A detailed comparison is shown in Table 3.

5 Conclusions

To protect human health, indoor environmental conditions need to be controlled. Hence, the development of an Arduino system with an integrated fuzzy logic controller is essential. The proposed system monitors the enclosed areas by settling the ventilation speed. This speed relies on the system's three inputs: temperature, humidity, and air quality. These sensors gather information from the surrounding area before it is processed to determine the proper ventilation speed. This

Table 3 Comparison between the proposed system and (Preethi and Tamilarasan 2021)

Technique	Hardware	Measurements	Analysis Method	Environment	System output
Preethi and Tamilarasan (2021)	DHT-22 MQ-7 Arduino UNO Smart LCD RF Module	Temperature AQI	I-ANFIS	Outdoors	The air pollution severities and its location
The proposed system	DHT-11 MQ-135 Arduino UNO DC Motor	Temperature Humidity AQI	Fuzzy logic	Indoors	A scaler value to control the speed of a ventilation mechanism system to reduce the air pollution

decision is made based on the set of rules in the fuzzy logic controller. The proposed system reduces the power consumption in comparison with the previous version because the ventilation mechanism operates at lower speeds when indoor environmental conditions are acceptable. Moreover, the new system has a larger range between the minimum and maximum values, which provides a better distribution for the output ventilation speeds. The proposed system is also compared to the technique reported in (Preethi and Tamilarasan 2021) to show the main differences between the two. In future work, the system can be improved in different ways. One possible improvement is the addition of more sensors for obtaining more accurate values for the environmental conditions. Another improvement might be to use several sensors of the same type in different places of the enclosed area to enable more accurate measurements. Moreover, adding more member functions for both inputs and the output of the fuzzy logic controller will increase the accuracy of the ventilation speed value.

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